Enhancing data annotation in ChEMBL for robust analysis

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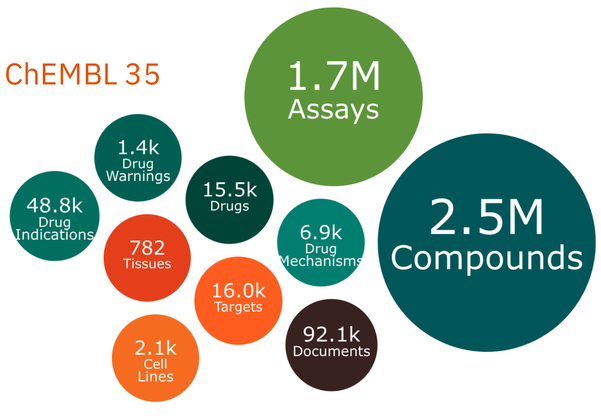
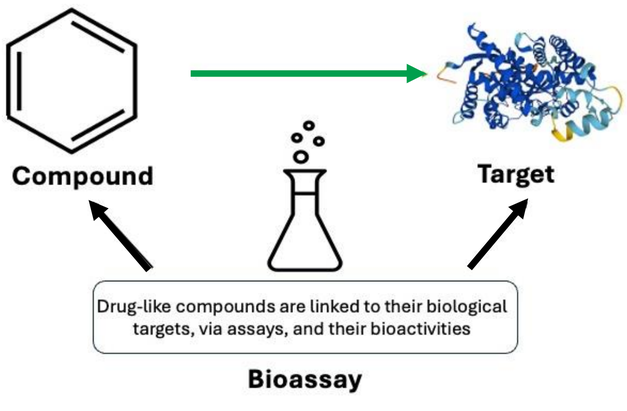
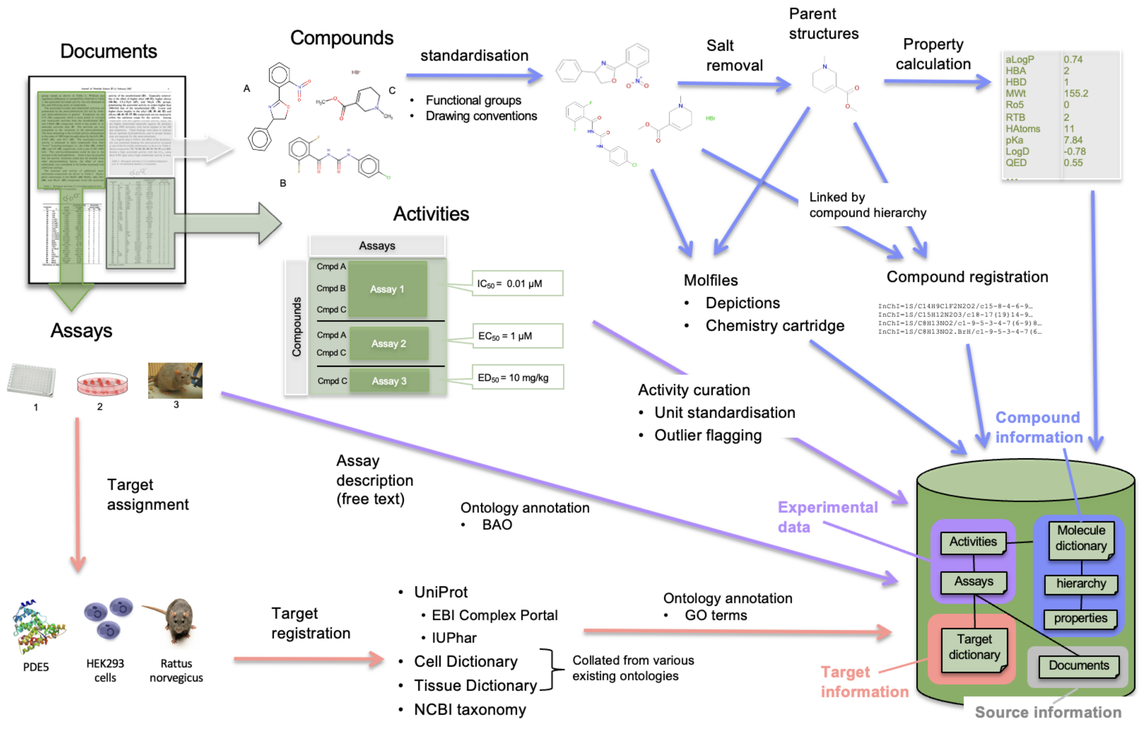
ChEMBL data

ChEMBL is a manually curated database of bioactive molecules with drug-like properties, bringing together chemical, bioactivity and genomic data to aid drug discovery. ChEMBL was named by the Global Biodata Coalition as a "Global Core Biodata Resource", highlighting the value of this curated resource to the drug discovery community and our need to be responsive to our user requirements.







The ChEMBL data ingestion and integration pipeline is evolving with our communities' needs, supported by a series of new data checks. Bioactivity data linking compounds, targets and bioassays is supported by diverse metadata fields allowing bespoke user queries. Value is added through existing manual and automated annotation, and projects in development which are exploring machine learning for further annotation.

Natural Language Processing (NLP) for improved assay annotation

As well as structured bioassay data, free-text is permitted in some fields to ensure ease and flexibility of deposition. To enhance value, we are exploring natural language processing methods to annotate broad assay categories and specific experimental methods.

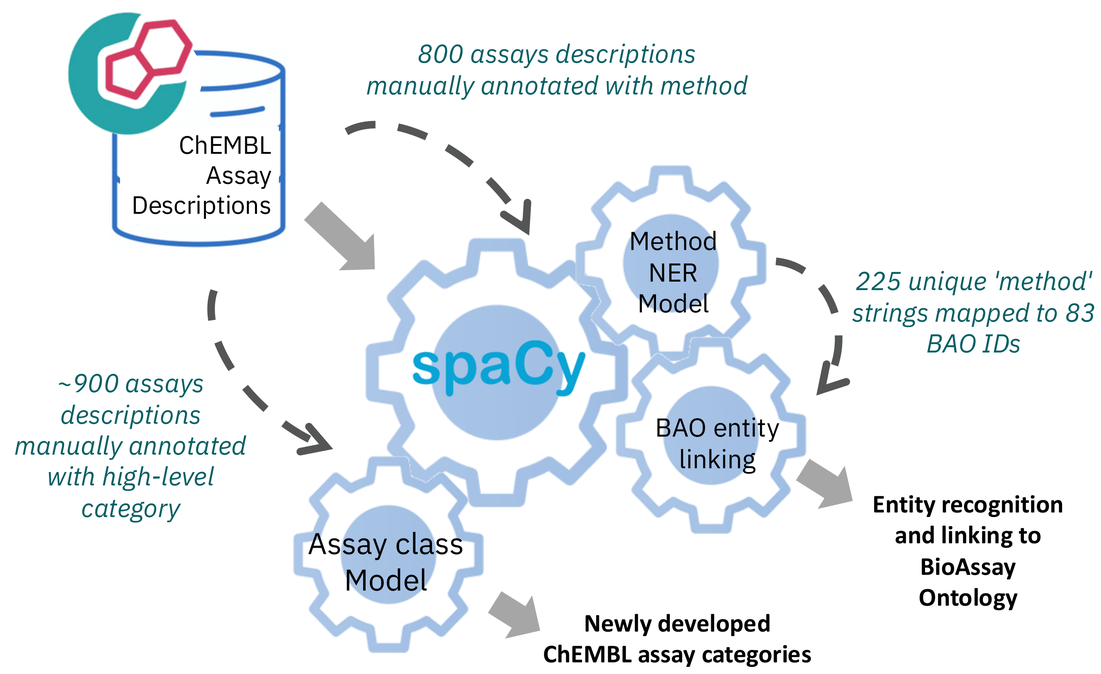
Assay categories will help users quickly access similar experiments. Linking experimental data to an ontology describing the assay method will enable users to better understand and more finely filter the data they are accessing.

A spaCy [1] multi-class text classification model was developed to annotate the largest resulting assay categories for Binding and Functional assays. Simultaneously, a spaCy Named Entity Recognition (NER) model was trained to identify the specific method entity within the ChEMBL assay descriptions. We are evaluating the text2term Python package [2] for linking method entities to the BioAssay Ontology (BAO) [3].

The models were run on ChEMBL 35 and 200 assays were randomly chosen for external validation by manual curation. For the broad category, 86% of the annotations agreed when considering predictions with a score > 0.7. The NER model had 89% agreement with the manually curated validation set.

Pipeline for assay annotations and assay categories

Initially focused on annotation of broad classification of overall assay class, and the experimental methods, which are not currently standardised to ontologies



Novel assay annotations and assay categories

Assay description

Agonist activity at GAL4 DNA binding domain fused glucocorticoid receptor ligand binding domain (unknown origin) at 0.25 to 25 uM by FRET-based assay **Experimental Method**

Named entity recognition	Ontology linking to BioAssay Ontology		Benefit
	BAO Preferred Term	BAO ID	
FRET-based assay	fluorescence resonance energy transfer	BAO_0000001	Specific annotation of assay features allows better comparison and grouping of assays for modelling. Current work focuses on experimental method, with future plans to expand other features e.g. setup, substrates, probes

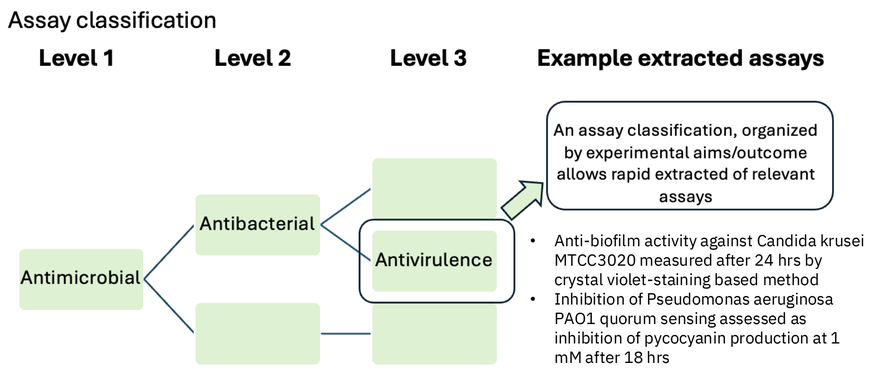
New assay category	Ontology linking to BioAssay Ontology		Benefit
	BAO Preferred Term	BAO ID	
Protein activity	functional target-based	BAO_0013016	Highest level of assay classification hierarchy, designed to replace the current assay types

Refer to BAO hierarchy

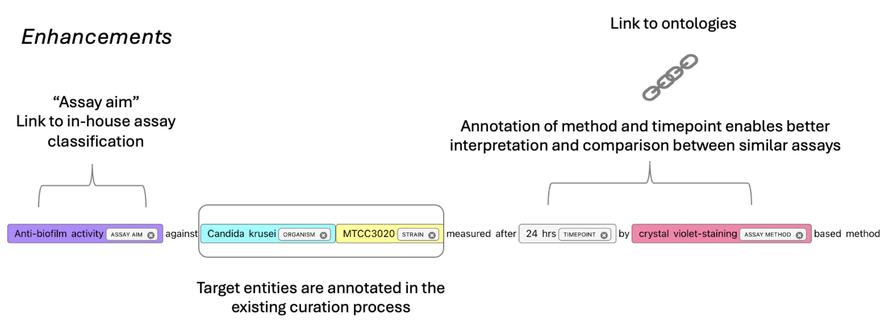
- physical detection method
  - biosensor method
  - fluorescence method
    - flow cytometry
    - fluorescence intensity
    - fluorescence polarization
    - **fluorescence resonance energy transfer**
    - thermal shift

Future goals and vision - Towards FAIRer bioassay data

Assay classification



Enhancements



Assay description

1) An assay classification to enhance **findability**

How can I find and extract all bacterial anti-virulence assays?

2) Enhanced annotation for **interoperable**, **reproducible** and **ML-ready data**

How can I interpret and compare my anti-virulence assay?

www.ebi.ac.uk

References
1. Matthew Honnibal, Ines Montani, Sofie Van Landeghem & Adriane Boyd. spaCy: Industrial-strength Natural Language Processing in Python. https://doi.org/10.5281/ZENODO.1212303 (2020).
2. Gonçalves, R. S. *et al.* The text2term tool to map free-text descriptions of biomedical terms to ontologies. Preprint at https://doi.org/10.48550/arXiv.2407.02626 (2024). https://github.com/rsgoncalves/text2term.
3. BioAssay Ontology. http://bioassayontology.org/

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